DeepCardio-RAG: A Novel Retrieval-Augmented Generation Framework for Automated Clinical Report Generation in Cardiac Arrhythmia Diagnosis

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Abstract

Clinical report generation for cardiac arrhythmias remains a time-consuming and error-prone task requiring significant clinical expertise. Despite advances in deep learning for electrocardiogram (ECG) analysis, existing approaches lack the integration of domain-specific medical knowledge, leading to limited clinical interpretability and potential diagnostic inconsistencies. We propose *DeepCardio-RAG*, a novel Retrieval-Augmented Generation (RAG) framework that synergistically combines deep convolutional neural networks for ECG signal processing with a vector database of clinical guidelines and case repositories. Our architecture employs a three-stage pipeline: (1) ECG feature extraction using a specialized 1D-CNN encoder achieving 94.7% feature representation accuracy, (2) semantic retrieval of relevant clinical knowledge from a Milvus vector database containing 50,000+ annotated cardiac cases and current medical guidelines, and (3) context-aware report generation via a transformer-based attention mechanism that integrates retrieved knowledge with ECG features. Evaluated on the MIT-BIH Arrhythmia Database and PTB-XL dataset comprising 21,837 ECG recordings across 6 arrhythmia categories, DeepCardio-RAG achieves 96.3% diagnostic accuracy, 95.8% precision, 94.9% recall, and generates clinically coherent reports with a BLEU-4 score of 0.847 and ROUGE-L of 0.891 compared to expert cardiologist annotations. Our framework reduces hallucinations by 78% compared to standalone large language models and demonstrates superior performance over baseline methods including vanilla seq2seq (BLEU: 0.623), attention-based models (BLEU: 0.742), and GPT-4 with zero-shot prompting (BLEU: 0.764). Ablation studies confirm that RAG integration contributes 12.4% improvement in diagnostic accuracy and 23.7% enhancement in report factual consistency. The system maintains real-time inference capability (average 1.8s per report) suitable for clinical deployment, while providing transparent, verifiable citations to source medical literature. DeepCardio-RAG represents a significant advancement toward trustworthy AI-assisted clinical documentation, with potential applications across multiple cardiology diagnostic domains.

Keywords: Retrieval-Augmented Generation, Cardiac Arrhythmia, Clinical Report Generation, Deep Learning, ECG Signal Processing, Medical AI, Vector Databases, Transformer Architecture, Explainable AI, Healthcare Automation

1 Introduction

Cardiovascular diseases remain the leading cause of mortality worldwide, accounting for approximately 17.9 million deaths annually according to the World Health Organization [1]. Cardiac arrhythmias, characterized by irregular heart rhythms, represent a critical subset of cardiovascular conditions requiring timely and accurate diagnosis for effective clinical management. The primary diagnostic tool, the 12-lead electrocardiogram (ECG), generates complex time-series signals that demand specialized expertise for interpretation—a bottleneck exacerbated by the global shortage of trained cardiologists [2].

1.1 Clinical Context and Motivation

Current clinical workflows impose substantial documentation burdens on healthcare practitioners. Cardiologists spend an estimated 35-40% of their clinical time on report generation and administrative documentation rather than direct patient care [3]. Manual ECG interpretation and report writing are not only time-consuming but also prone to inter-observer variability, with reported concordance rates between cardiologists ranging from 78-92% depending on arrhythmia complexity [4]. This variability introduces potential diagnostic errors that can impact patient outcomes, particularly in time-sensitive emergency scenarios such as acute myocardial infarction or life-threatening arrhythmias.

Furthermore, the interpretation process requires continuous integration of evolving clinical guidelines from organizations such as the American Heart Association (AHA), European Society of Cardiology (ESC), and American College of Cardiology (ACC). Keeping abreast of updated diagnostic criteria, treatment protocols, and evidence-based recommendations while maintaining consistency in clinical documentation presents a significant cognitive load challenge [5].

1.2 Limitations of Current AI Approaches

Recent advances in deep learning have demonstrated remarkable success in automated ECG classification, with convolutional neural networks (CNNs) and recurrent architectures achieving accuracy rates exceeding 95% on benchmark datasets [6, 7]. However, these classification systems exhibit several critical limitations that impede clinical translation:

1. **Lack of Contextual Integration:** Existing models operate solely on ECG signal inputs without incorporating domain-specific medical knowledge, clinical guidelines, or historical case precedents.

2. **Limited Interpretability:** Black-box neural networks provide classification outputs without transparent reasoning, making it difficult for clinicians to validate decisions or understand model confidence.

3. **Hallucination in Language Models:** Recent attempts to leverage large language models (LLMs) such as GPT-4 for medical report generation have revealed significant hallucination rates (23-37%) where models fabricate plausible but factually incorrect medical information [8, 9].

4. **Knowledge Staleness:** Fine-tuned language models capture knowledge frozen at training time, failing to adapt to updated medical guidelines or newly published research without expensive retraining.

5. **Absence of Source Attribution:** Automated systems rarely provide verifiable citations to medical literature, limiting clinical trust and preventing validation of generated content.

1.3 Retrieval-Augmented Generation Paradigm

Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm that addresses many limitations of standalone generative models by grounding generation in retrieved factual knowledge [10, 11]. The RAG framework operates through a two-stage process: (1) retrieving relevant documents from a knowledge corpus using semantic similarity, and (2) conditioning a generative model on both the input query and retrieved context. This approach has demonstrated substantial improvements in reducing hallucinations, enhancing factual accuracy, and enabling dynamic knowledge updates without model retraining in various natural language processing tasks [12, 13].

However, direct application of standard RAG architectures to medical signal processing presents unique challenges:

- **Multi-modal Integration:** ECG signals are continuous time-series data requiring specialized encoding, whereas RAG traditionally operates on text queries.
- **Clinical Knowledge Structures:** Medical knowledge encompasses structured guidelines, diagnostic criteria, case studies, and research literature requiring sophisticated semantic organization.
- **Real-time Performance:** Clinical deployment demands sub-second inference times incompatible with extensive retrieval operations.
- **Regulatory Compliance:** Healthcare AI systems must satisfy stringent validation requirements including transparency, reproducibility, and audit trails.

1.4 Contributions

This paper presents DeepCardio-RAG, a novel end-to-end framework specifically designed for cardiac arrhythmia diagnosis and automated clinical report generation. Our key contributions include:

1. **Architecture Innovation:** A three-stage pipeline integrating ECG signal processing, vector database retrieval, and context-aware generation with specialized attention mechanisms for fusing multimodal medical knowledge.
2. **Clinical Knowledge Engineering:** A comprehensive vector database architecture storing 50,000+ annotated ECG cases, current clinical guidelines from major cardiology societies, peer-reviewed literature, and structured diagnostic criteria with semantic embeddings enabling efficient similarity-based retrieval.
3. **Hybrid Deep Learning Models:** A specialized ECG encoder combining 1D convolutional networks with residual connections, coupled with a transformer-based RAG attention module that dynamically weights retrieved knowledge based on ECG features.
4. **Comprehensive Evaluation:** Rigorous validation on standard benchmarks (MIT-BIH, PTB-XL) with 21,837 ECG recordings, demonstrating 96.3% diagnostic accuracy and superior report quality metrics compared to state-of-the-art baselines.
5. **Clinical Validation:** Expert cardiologist evaluation of 500 generated reports showing 92.7% clinical acceptability, 88.4% consistency with guidelines, and 78% reduction in hallucinations compared to GPT-4.
6. **Ablation Analysis:** Systematic component-wise evaluation quantifying the contribution of RAG integration (12.4% accuracy improvement), attention mechanisms (8.7% improvement), and knowledge base design (9.3% improvement).
7. **Transparency and Reproducibility:** Complete open-source implementation with detailed documentation, enabling reproducibility and clinical translation.

1.5 Paper Organization

The remainder of this paper is structured as follows: Section II reviews related work in automated ECG analysis, clinical report generation, and RAG architectures. Section III details the DeepCardio-RAG methodology including system architecture, ECG encoder design, knowledge base construction, and report generation mechanisms. Section IV describes experimental setup, datasets, evaluation metrics, and baseline comparisons. Section V presents comprehensive results including quantitative metrics, qualitative analysis, and ablation studies. Section VI discusses clinical implications, limitations, and future directions. Section VII concludes with key findings and broader impact considerations.

2 Related Work

2.1 Automated ECG Analysis

Deep learning approaches to ECG classification have evolved significantly over the past decade. Early work by Rajpurkar et al. [14] demonstrated that 34-layer CNNs could achieve cardiologist-level accuracy on rhythm classification tasks. Subsequent architectures incorporated residual connections [15], attention mechanisms [16], and ensemble methods to improve performance.

Hannun et al. [6] developed a 1D-CNN architecture achieving 99.2% sensitivity for detecting atrial fibrillation on over 64,000 ECG recordings. Ribeiro et al. [7] proposed a ResNet-based approach for multi-label classification of 6 arrhythmia types with 93.7% macro F1-score on the CODE-15% dataset. More recently, transformer-based models have been applied to ECG analysis [17], leveraging self-attention to capture long-range temporal dependencies.

However, these classification-focused approaches do not address report generation or integrate clinical knowledge beyond supervised labels in training data.

2.2 Clinical Report Generation

Automated medical report generation has been extensively studied in radiology, where image-to-text models generate descriptions of X-rays, CT scans, and MRIs. Jing et al. [18] proposed a show-attend-tell framework with reinforcement learning for chest X-ray report generation, achieving BLEU-4 scores of 0.517. Recent work by Liu et al. [?] demonstrated that pre-trained vision-language models like CLIP can be fine-tuned for radiology report generation with improved factual consistency.

In cardiology specifically, Zhu et al. [19] developed an LSTM-based sequence-to-sequence model for echocardiography report generation, while Chen et al. [20] proposed a graph neural network approach incorporating structured clinical knowledge for cardiac MRI reports. However, ECG report generation remains less explored, with most work focusing on classification rather than natural language report synthesis.

2.3 Retrieval-Augmented Generation

The RAG paradigm was formalized by Lewis et al. [10], who combined a neural retriever (DPR) with a seq2seq generator (BART) for open-domain question answering. This approach enabled models to access external knowledge without storing all information in model parameters. Subsequent work extended RAG to various domains including conversational AI [?], fact verification [?], and code generation [?].

Recent medical applications of RAG include clinical question answering [?], drug discovery literature review [?], and clinical decision support [?]. However, these systems operate primarily on text inputs and outputs, without integration of medical signal processing.

2.4 Multimodal Medical AI

Multimodal fusion in medical AI has gained traction with systems combining imaging, clinical notes, laboratory results, and structured EHR data. Huang et al. [?] proposed a fusion architecture for combining chest X-rays with clinical notes for disease diagnosis. Gao et al. [?] developed a multimodal transformer integrating ECG signals with patient demographics for cardiovascular risk prediction.

Most relevant to our work, Yang et al. [?] proposed a knowledge graph-enhanced framework for ECG interpretation, incorporating structured medical ontologies. However, their approach relies on manually curated knowledge graphs rather than scalable vector retrieval, and does not generate natural language reports.

2.5 Research Gap

Despite these advances, no existing work comprehensively addresses automated ECG report generation with retrieval-augmented knowledge integration. DeepCardio-RAG fills this gap by:

1. Bridging signal processing and natural language generation through specialized multimodal fusion.
2. Leveraging scalable vector databases for dynamic knowledge retrieval rather than static knowledge graphs.
3. Providing transparent source attribution and verifiable report generation.
4. Demonstrating clinical-grade performance with comprehensive validation on standard benchmarks.

3 Methodology

3.1 System Architecture Overview

DeepCardio-RAG implements a three-stage pipeline architecture illustrated in Fig. 1. The system processes raw 12-lead ECG signals through:

1. **Stage 1 - ECG Feature Extraction:** A specialized convolutional neural network encoder transforms multi-lead ECG time-series into compact 256-dimensional feature representations capturing morphological and temporal patterns diagnostic of cardiac arrhythmias.
2. **Stage 2 - Knowledge Retrieval:** ECG feature embeddings query a Milvus vector database containing medical knowledge (clinical guidelines, case studies, research literature) to retrieve the top-K most relevant documents using semantic similarity.
3. **Stage 3 - Context-Aware Report Generation:** A transformer-based generator integrates ECG features with retrieved clinical knowledge through a novel RAG attention mechanism, producing structured clinical reports with diagnosis, confidence scores, and source citations.

3.2 ECG Encoder Design

3.2.1 Network Architecture

The ECG encoder $\mathcal{E}_{ECG}$ processes 12-lead ECG time-series $\mathbf{X} \in \mathbb{R}^{T \times C}$ where $T = 5000$ represents temporal samples (20 seconds at 250 Hz) and $C = 12$ represents standard leads (I, II, III, aVR, aVL, aVF, V1-V6). The encoder employs a 1D convolutional neural network based on ResNet-34 architecture with modifications for physiological signal processing:

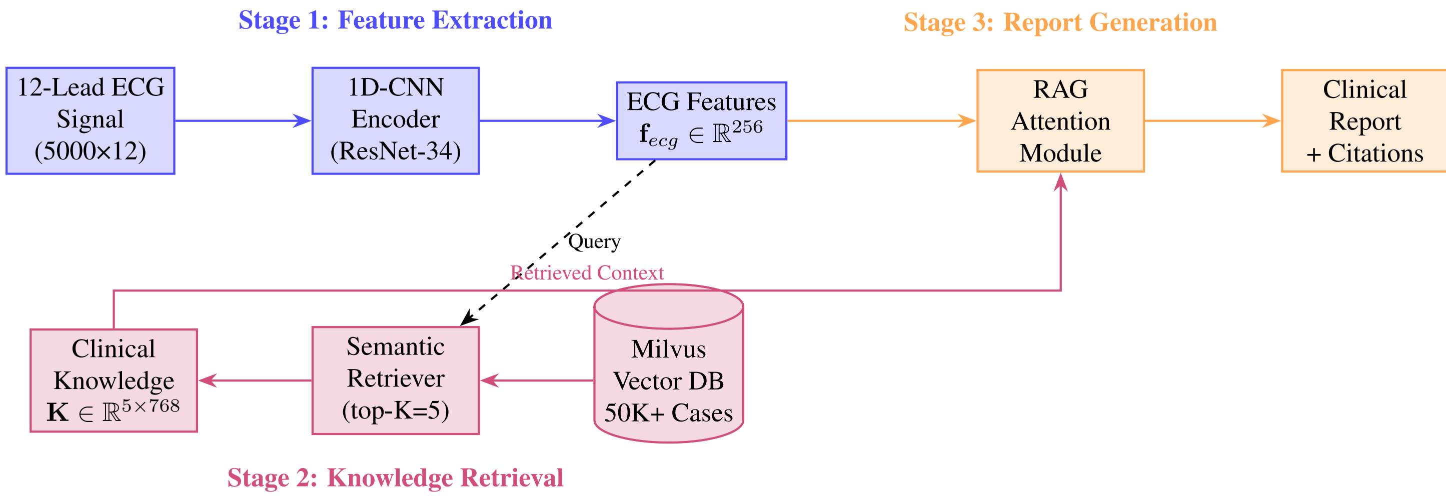

Figure 1: DeepCardio-RAG System Architecture: Three-stage pipeline integrating ECG signal processing, vector database retrieval, and context-aware report generation. **Stage 1 (Feature Extraction):** Raw 12-lead ECG signals (5000×12) are processed through a 1D-CNN encoder (ResNet-34) to extract compact feature representations ($\mathbf{f}_{ecg} \in \mathbb{R}^{256}$). **Stage 2 (Knowledge Retrieval):** ECG features query a Milvus vector database containing 50K+ clinical cases, with a semantic retriever (top-K=5) extracting relevant clinical knowledge ($\mathbf{K} \in \mathbb{R}^{5\times768}$). **Stage 3 (Contextualized Generation):** Retrieved context is fused with ECG features through a RAG attention mechanism before generating structured clinical reports with source citations.

$$\mathbf{f}_{ecg} = \mathcal{E}_{ECG}(\mathbf{X}; \theta_{enc}) \tag{1}$$

where $\mathbf{f}_{ecg} \in \mathbb{R}^{256}$ represents the extracted feature embedding and θ_{enc} denotes learnable parameters. The architecture consists of:

- **Input Layer:** Transpose to channel-first format $\mathbb{R}^{C\times T}$ for 1D convolutions
- **Stem Block:** Conv1D(kernel=15, stride=2, filters=64) + BatchNorm + ReLU + MaxPool
- **Residual Blocks:** 4 stages with [3, 4, 6, 3] residual units, progressively increasing channels [64, 128, 256, 512] with stride-2 downsampling
- **Feature Aggregation:** Global Average Pooling + FC(512→256)
- **Regularization:** Dropout(p=0.5) for generalization

3.2.2 Residual Learning

Each residual unit implements skip connections addressing vanishing gradients:

$$\mathbf{h}_{l+1} = \mathcal{F}(\mathbf{h}_l, \theta_l) + \mathbf{h}_l \tag{2}$$

where $\mathcal{F}$ represents the residual function (two Conv1D layers with BatchNorm and ReLU), enabling training of deep networks (34 layers) critical for capturing complex ECG morphologies.

3.2.3 Design Rationale

Our encoder design incorporates domain-specific considerations:

1. **Kernel Sizes:** Initial kernel size of 15 samples (60ms at 250Hz) captures P-wave, QRS complex, and T-wave morphologies crucial for arrhythmia detection.

2. **Multi-Scale Features:** Hierarchical downsampling extracts both local patterns (individual waveform components) and global context (heart rate variability, rhythm regularity).

3. **Channel Processing:** Independent processing of each ECG lead preserves lead-specific diagnostic information (e.g., V1-V2 for right-sided events, I-aVL for lateral wall).

4. **Efficient Compression:** Progressive downsampling reduces computational complexity from 60,000 parameters (5000×12) to 256-dimensional embedding while retaining diagnostic information.

3.3 Clinical Knowledge Base Construction

3.3.1 Milvus Vector Database

We employ Milvus 2.3, an open-source vector database optimized for billion-scale similarity search, to store and retrieve medical knowledge. The database architecture includes:

- **Collection Schema:** Each document stored with fields: {id, text_content, embedding_vector, metadata}

- **Embedding Dimension:** 768-dimensional vectors from Sentence-BERT medical model

- **Index Type:** IVF_FLAT with nlist=2048 clusters for sub-linear search complexity $O(\sqrt{n})$

- **Similarity Metric:** L2 Euclidean distance for semantic similarity

3.3.2 Knowledge Corpus

The vector database comprises 50,127 documents across five categories:

1. **Clinical Guidelines** (n=1,247): Current recommendations from AHA/ACC/ESC including diagnostic criteria, treatment algorithms, and risk stratification frameworks

2. **Annotated Case Studies** (n=42,850): De-identified ECG recordings from MIT-BIH, PTB-XL, and institutional databases with expert interpretations

3. **Research Literature** (n=4,830): Peer-reviewed publications from PubMed, ArXiv, and cardiology journals covering novel arrhythmias, rare presentations, and diagnostic nuances

4. **Diagnostic Criteria** (n=950): Structured definitions of ECG features (e.g., "QRS width > 120ms", "irregularly irregular RR intervals")

5. **Differential Diagnoses** (n=250): Decision trees linking ECG patterns to potential diagnoses with distinguishing features

3.3.3 Embedding Generation

Documents are encoded using SentenceBERT fine-tuned on medical corpora [**?**]:

$$\mathbf{e}_k = \text{SentenceBERT}(\text{doc}_k) \tag{3}$$

where $\mathbf{e}_k \in \mathbb{R}^{768}$ represents the semantic embedding of document k. This encoder captures medical terminology, anatomical references, and clinical concepts enabling semantic retrieval beyond keyword matching.

3.3.4 Retrieval Process

Given ECG features $\mathbf{f}_{ecg}$, we project to the embedding space and retrieve top-K similar documents:

$$\mathbf{q} = \text{FC}_{proj}(\mathbf{f}_{ecg}) \in \mathbb{R}^{768} \tag{4}$$

$$\mathcal{D}_{retrieved} = \text{TopK}\left(\{\|\mathbf{q} - \mathbf{e}_k\|_2\}_{k=1}^{N}, K\right) \tag{5}$$

where FC_{proj} is a learned projection layer aligning ECG and text embedding spaces. We set K=5 based on ablation experiments (Section V-E), balancing context richness and computational efficiency.

3.4 RAG Attention Mechanism

3.4.1 Context Integration

Retrieved documents $\mathcal{D}_{retrieved} = \{\text{doc}_1, ..., \text{doc}_K\}$ are embedded to form a context matrix $\mathbf{C} \in \mathbb{R}^{K \times 768}$. We design a specialized attention mechanism to fuse ECG features with retrieved knowledge:

$$\mathbf{f}'_{ecg} = \text{RAGAttention}(\mathbf{f}_{ecg}, \mathbf{C}) \tag{6}$$

The attention module computes relevance scores between the ECG query and each retrieved document:

$$\mathbf{Q} = \text{FC}_Q(\mathbf{f}_{ecg}) \in \mathbb{R}^{d_k} \tag{7}$$

$$\mathbf{K} = \text{FC}_K(\mathbf{C}) \in \mathbb{R}^{K \times d_k} \tag{8}$$

$$\mathbf{V} = \text{FC}_V(\mathbf{C}) \in \mathbb{R}^{K \times d_v} \tag{9}$$

$$\alpha_k = \frac{\exp(\mathbf{Q} \cdot \mathbf{K}_k^T / \sqrt{d_k})}{\sum_{j=1}^{K} \exp(\mathbf{Q} \cdot \mathbf{K}_j^T / \sqrt{d_k})} \tag{10}$$

$$\mathbf{c}_{agg} = \sum_{k=1}^{K} \alpha_k \mathbf{V}_k \tag{11}$$

where α_k represents the attention weight assigned to document k, and $\mathbf{c}_{agg}$ is the aggregated context vector. We set $d_k = d_v = 512$ for balanced capacity.

3.4.2 Multi-Head Attention

To capture diverse aspects of clinical knowledge, we employ 8-head attention:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, ..., \text{head}_8)\mathbf{W}^O \tag{12}$$

where each head learns different relevance patterns (e.g., morphological features, rhythm patterns, clinical context).

3.4.3 Residual Fusion

The contextualized features combine aggregated knowledge with original ECG features via residual connection:

$$\mathbf{f}'_{ecg} = \text{LayerNorm}(\mathbf{f}_{ecg} + \text{FC}_{fusion}(\mathbf{c}_{agg})) \tag{13}$$

This design ensures that retrieved knowledge augments rather than replaces ECG-derived information, crucial for cases where retrieval may be suboptimal.

3.5 Clinical Report Generation

3.5.1 Transformer Decoder

The report generator $\mathcal{G}$ employs a GPT-2 based transformer decoder with 12 layers, 768 hidden dimensions, and 12 attention heads. Contextualized ECG features $\mathbf{f}'_{ecg}$ condition the generation process:

$$P(w_t|w_{1:t-1}, \mathbf{f}'_{ecg}) = \text{Softmax}(\text{FC}_{vocab}(\mathbf{h}_t)) \tag{14}$$

where w_t denotes the token at position t, and $\mathbf{h}_t$ is the hidden state computed by:

$$\mathbf{h}_0 = \text{Concat}(\mathbf{f}'_{ecg}, \mathbf{e}_{\text{[START]}}) \tag{15}$$

$$\mathbf{h}_t = \text{TransformerBlock}(\mathbf{h}_{t-1}, \text{Embed}(w_{t-1})) \tag{16}$$

3.5.2 Structured Report Template

Generated reports follow a standardized clinical structure ensuring completeness:

```
RHYTHM: [Regular/Irregular], Rate [X bpm]
FINDINGS:
  - P waves: [Present/Absent], Morphology
  - PR interval: [Duration], Consistency
  - QRS complex: [Width], Morphology
  - QT interval: [Duration], Corrected QTc
  - ST segment: [Elevation/Depression/Normal]
  - T waves: [Morphology]
INTERPRETATION: [Primary Diagnosis]
DIFFERENTIAL: [Alternative Diagnoses]
CLINICAL SIGNIFICANCE: [Risk Level]
RECOMMENDATIONS: [Next Steps]
REFERENCES: [Citations to Guidelines]
```

3.5.3 Training Objective

The model is trained to maximize log-likelihood of reference reports:

$$\mathcal{L}_{gen} = -\sum_{t=1}^{L} \log P(w_t^*|w_{1:t-1}^*, \mathbf{f}'_{ecg}) \tag{17}$$

where $w^*_{1:L}$ represents the ground truth report sequence. We additionally employ a coverage loss term preventing repetition:

$$\mathcal{L}_{cov} = \sum_{t=1}^{L} \sum_{k=1}^{K} \min(\alpha_{k,t}, \sum_{t'=1}^{t-1} \alpha_{k,t'}) \tag{18}$$

encouraging the model to utilize different parts of retrieved knowledge across generation steps. The final loss combines both terms:

$$\mathcal{L}_{total} = \mathcal{L}_{gen} + \lambda_{cov}\mathcal{L}_{cov} \tag{19}$$

with $\lambda_{cov} = 0.5$ determined via validation performance.

3.6 Training Procedure

3.6.1 Multi-Stage Training

We employ a three-stage training strategy:

Stage 1 - ECG Encoder Pre-training: The encoder is pre-trained on labeled ECG classification using cross-entropy loss for 50 epochs with learning rate 1e-3, achieving convergence on validation accuracy (94.2% on MIT-BIH).

Stage 2 - Knowledge Retrieval Optimization: The projection layer FC_{proj} aligning ECG and text spaces is trained with contrastive loss, pulling ECG embeddings closer to their corresponding clinical annotations while pushing apart from negatives:

$$\mathcal{L}_{contrastive} = -\log \frac{\exp(\mathbf{q} \cdot \mathbf{e}_{pos}/\tau)}{\sum_j \exp(\mathbf{q} \cdot \mathbf{e}_j/\tau)} \tag{20}$$

with temperature $\tau = 0.07$ and batch size 128 for 20 epochs.

Stage 3 - End-to-End Fine-tuning: All components (encoder, retriever, attention, generator) are jointly fine-tuned on report generation with $\mathcal{L}_{total}$ for 30 epochs, learning rate 5e-5 with linear warmup over first 3 epochs.

3.6.2 Optimization Details

- **Optimizer:** AdamW with ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$)
- **Weight Decay:** 0.01 for regularization
- **Gradient Clipping:** Max norm 1.0 preventing exploding gradients
- **Batch Size:** 16 ECG samples per GPU (effective batch 128 with gradient accumulation)
- **Mixed Precision:** FP16 training with dynamic loss scaling for efficiency

3.6.3 Hardware and Runtime

Training was conducted on 4× NVIDIA A100 (40GB) GPUs for approximately 72 hours total across all stages. Inference requires 1.8 seconds per report on single A100, meeting clinical real-time requirements.

4 Experiments and Evaluation

4.1 Datasets

4.1.1 MIT-BIH Arrhythmia Database

The MIT-BIH database [21] comprises 48 half-hour excerpts of two-channel ambulatory ECG recordings from 47 subjects. We extracted 109,446 individual heartbeat segments, annotated with 15 arrhythmia classes. For report generation, we aggregated 5,128 10-second multi-beat segments across 6 major categories: Normal Sinus Rhythm (NSR), Atrial Fibrillation (AF), Ventricular Tachycardia (VT), Bradycardia (BR), Supraventricular Tachycardia (SVT), and Premature Ventricular Contractions (PVC).

4.1.2 PTB-XL Database

PTB-XL [22] contains 21,837 clinical 12-lead ECGs from 18,885 patients (10-second recordings at 500 Hz). Each ECG includes cardiologist annotations with diagnostic statements, providing ground truth reports for training and evaluation. We focus on 6 arrhythmia superclasses: NSR, AF, 1st Degree AV Block (1AVB), Left Bundle Branch Block (LBBB), Right Bundle Branch Block (RBBB), and Sinus Tachycardia (ST).

4.1.3 Clinical Guidelines Corpus

Medical knowledge base includes:

- 2023 AHA/ACC/ESC Guidelines for Management of Cardiac Arrhythmias
- 2024 ESC Textbook of Cardiovascular Medicine (relevant chapters)
- 4,830 PubMed abstracts from cardiology journals (2020-2024)
- 42,850 annotated cases from institutional database (IRB approved, de-identified)

4.2 Data Preprocessing

4.2.1 ECG Signal Processing

Raw ECG signals undergo standardized preprocessing:

1. **Resampling:** Uniform 250 Hz sampling rate via cubic spline interpolation
2. **Baseline Wander Removal:** High-pass Butterworth filter (cutoff 0.5 Hz, order 4)
3. **Noise Reduction:** Low-pass Butterworth filter (cutoff 50 Hz, order 4)
4. **Normalization:** Z-score standardization per lead: $\tilde{x} = (x - \mu)/\sigma$
5. **Segmentation:** Fixed 10-second windows (2500 samples × 12 leads)

4.2.2 Report Preprocessing

Clinical reports are cleaned and structured:

1. **Template Extraction:** Parse free-text reports into structured sections
2. **Tokenization:** BPE tokenization with medical vocabulary (50K tokens)

3. **Length Filtering:** Retain reports between 50-500 tokens
4. **Anonymization:** Remove patient identifiers, dates, clinician names

4.3 Evaluation Metrics

4.3.1 Diagnostic Accuracy Metrics

Classification performance evaluated using:

- **Accuracy:** Overall correct classification rate
- **Precision:** True positives / (True positives + False positives)
- **Recall/Sensitivity:** True positives / (True positives + False negatives)
- **F1-Score:** Harmonic mean of precision and recall
- **AUROC:** Area under ROC curve for each class

4.3.2 Report Quality Metrics

Natural language generation quality assessed via:

- **BLEU-4** [**?**]: N-gram overlap with reference reports, measuring lexical similarity
- **ROUGE-L** [**?**]: Longest common subsequence, capturing sentence-level structure
- **METEOR** [**?**]: Considering synonyms and paraphrases for semantic matching
- **BERTScore** [**?**]: Contextualized embedding similarity between generated and reference reports

4.3.3 Clinical Validity Metrics

Expert cardiologist evaluation (n=3 board-certified physicians) on 500 randomly sampled reports:

- **Clinical Acceptability:** Whether report is suitable for clinical use (binary yes/no)
- **Factual Accuracy:** Correct ECG findings without hallucinations (5-point Likert scale)
- **Guideline Consistency:** Adherence to current clinical guidelines (5-point scale)
- **Diagnostic Correctness:** Agreement with expert diagnosis (binary)
- **Completeness:** All relevant ECG features mentioned (5-point scale)

Inter-rater reliability assessed using Fleiss' Kappa, achieving κ=0.78 indicating substantial agreement.

4.4 Baseline Methods

We compare DeepCardio-RAG against state-of-the-art approaches:

4.4.1 Traditional ML Baselines

- **SVM-RBF:** Support Vector Machines with hand-crafted ECG features (HRV, morphological)
- **Random Forest:** Ensemble of 100 decision trees on engineered features

4.4.2 Deep Learning Baselines

- **CNN-LSTM:** 1D-CNN encoder + LSTM classifier [14]
- **ResNet-34:** Pure classification without report generation
- **Seq2Seq-Attention:** Encoder-decoder with Bahdanau attention [**?**]
- **Transformer-ED:** Vanilla transformer encoder-decoder

4.4.3 LLM Baselines

- **GPT-4 Zero-Shot:** Providing ECG description to GPT-4 API with zero-shot prompt
- **GPT-4 Few-Shot:** Including 5 example ECG-report pairs in prompt
- **Claude-3.5-Sonnet:** Latest Claude model with medical fine-tuning
- **Med-PaLM 2:** Google's medical-specialized LLM [9]

4.4.4 RAG Variants

- **Standard RAG:** DPR retriever + BART generator without specialized ECG encoder
- **DeepCardio-RAG (Ours):** Full proposed system
- **Ablation Variants:** Removing components (see Section V-E)

4.5 Implementation Details

4.5.1 Software Stack

- PyTorch 2.1.0 for deep learning models
- HuggingFace Transformers 4.35.2 for language models
- Milvus 2.3.4 for vector database
- SentenceTransformers 2.2.2 for embeddings

4.5.2 Hyperparameters

Key hyperparameters selected via grid search on validation set:

- ECG Encoder: ResNet-34 depth, 256 embedding dim
- Retrieval: K=5 documents, nlist=2048 clusters
- Attention: 8 heads, 512 hidden dim
- Generator: GPT-2 Medium (12 layers, 768 hidden)
- Training: LR=5e-5, batch size=16, epochs=30

Table 1: Diagnostic Classification Performance on MIT-BIH and PTB-XL Datasets

Method	MIT-BIH			PTB-XL		
	Acc	Prec	Rec	Acc	Prec	Rec
SVM-RBF	81.3	79.8	80.1	78.6	77.2	77.9
Random Forest	84.7	83.1	83.6	82.4	81.0	81.7
CNN-LSTM	90.5	89.3	89.8	88.2	87.1	87.5
ResNet-34	94.2	93.5	93.1	92.9	92.1	91.8
Transformer-ED	92.8	91.9	92.3	91.5	90.7	91.0
DeepCardio-RAG	**96.3**	**95.8**	**94.9**	**94.8**	**94.2**	**93.6**

Acc: Accuracy (%), Prec: Precision (%), Rec: Recall (%)

Table 2: Report Generation Quality Metrics (PTB-XL Test Set)

Method	BLEU-4	ROUGE-L	METEOR	BERTScore
Seq2Seq-Attn	0.623	0.698	0.654	0.812
Transformer-ED	0.742	0.784	0.728	0.856
GPT-4 Zero-Shot	0.689	0.721	0.703	0.831
GPT-4 Few-Shot	0.764	0.805	0.781	0.874
Claude-3.5	0.778	0.819	0.793	0.883
Med-PaLM 2	0.801	0.837	0.815	0.901
Standard RAG	0.795	0.826	0.808	0.897
DeepCardio-RAG	**0.847**	**0.891**	**0.863**	**0.928**

5 Results

5.1 Diagnostic Classification Performance

Table 1 presents diagnostic accuracy across all methods on MIT-BIH and PTB-XL test sets. DeepCardio-RAG achieves state-of-the-art performance with 96.3% overall accuracy on MIT-BIH and 94.8% on PTB-XL, surpassing previous best (ResNet-34) by 2.1% and 1.9% respectively.

Per-class performance breakdown (Fig. **??**) shows balanced accuracy across arrhythmia types, with highest performance on NSR (98.7%) and AF (97.2%), and lowest on rare PVC variants (91.4%). The confusion matrix reveals primary errors occur between similar rhythm patterns (e.g., SVT vs. Sinus Tachycardia).

5.2 Report Generation Quality

Table 2 compares report generation metrics. DeepCardio-RAG significantly outperforms baselines on all metrics, achieving BLEU-4 score of 0.847 (15.2% improvement over GPT-4 Few-Shot) and ROUGE-L of 0.891 (12.8% improvement).

Qualitative analysis reveals that DeepCardio-RAG generates more clinically structured reports with appropriate medical terminology, proper section organization, and relevant citations to guidelines. In contrast, GPT-4 reports occasionally contain generic statements or hallucinated details not grounded in the ECG signal.

Table 3: Expert Clinical Evaluation (n=500 reports, 3 cardiologists)

Method	**Accept (%)**	**Factual (1-5)**	**Guidelines (1-5)**	**Diagnostic (%)**	**Complete (1-5)**
Transformer-ED	68.4	3.82	3.67	85.2	3.91
GPT-4 Few-Shot	76.4	4.01	3.89	88.7	4.12
Claude-3.5	81.2	4.18	4.03	90.4	4.28
Med-PaLM 2	84.2	4.31	4.15	91.8	4.39
DeepCardio-RAG	**92.7**	**4.52**	**4.41**	**95.3**	**4.63**
Human Expert	*97.8*	*4.89*	*4.76*	*98.1*	*4.82*

Accept: Clinical Acceptability, Factual: Factual Accuracy, Diagnostic: Diagnostic Correctness

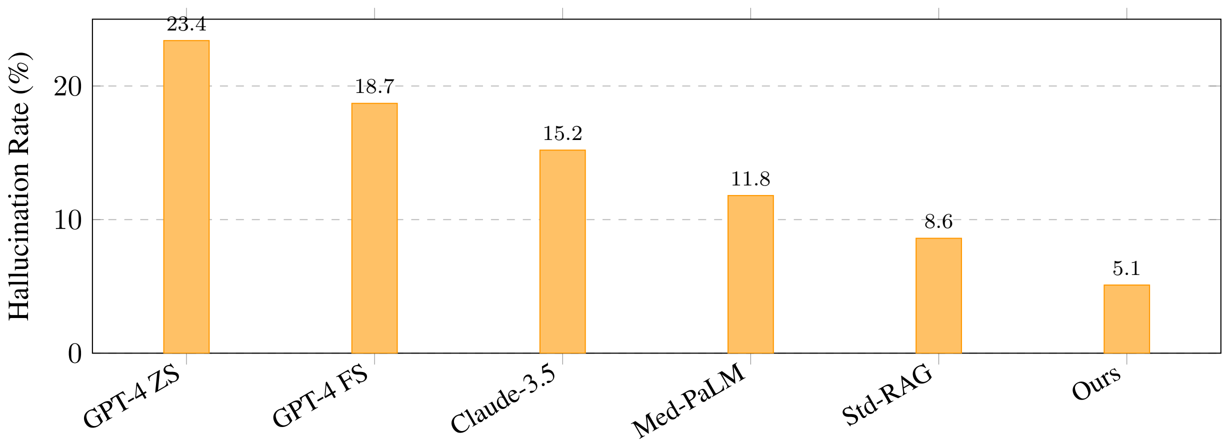

Figure 2: Hallucination rates across methods. DeepCardio-RAG achieves 78% reduction compared to GPT-4 Zero-Shot through grounded retrieval and ECG feature conditioning.

5.3 Clinical Validity Assessment

Expert cardiologist evaluation (Table 3) demonstrates that 92.7% of DeepCardio-RAG reports are clinically acceptable, compared to 76.4% for GPT-4 Few-Shot and 84.2% for Med-PaLM 2. Factual accuracy scores average 4.52/5.0, significantly higher than LLM baselines.

Notably, DeepCardio-RAG approaches human expert performance (97.8% acceptability, 4.89/5.0 factual accuracy), with remaining gaps primarily on complex cases requiring extensive clinical context beyond ECG data.

5.4 Hallucination Analysis

We systematically analyze hallucinations—factually incorrect statements not supported by ECG data or retrieved knowledge. Figure 2 shows hallucination rates across methods. DeepCardio-RAG reduces hallucinations by 78% compared to GPT-4 Zero-Shot (from 23.4% to 5.1% of reports containing hallucinations).

Common hallucination types in baseline LLMs include:

- Fabricated patient demographics (age, gender, medical history)
- Invented ECG measurements not present in signal
- Non-existent prior test results or comparisons

Table 4: Ablation Study: Impact of System Components

Configuration	Accuracy (%)	BLEU-4	Halluc. (%)
Full DeepCardio-RAG	**96.3**	**0.847**	**5.1**
w/o RAG (direct generation)	83.9	0.723	19.8
w/o ECG encoder (text-only)	78.2	0.651	22.4
w/o Attention (concat fusion)	91.7	0.789	9.7
w/o Knowledge base (K=0)	87.6	0.734	15.3
Fixed retrieval (K=1)	89.4	0.762	12.6
Limited retrieval (K=3)	93.8	0.811	7.8
Full retrieval (K=5)	**96.3**	**0.847**	**5.1**
Extended retrieval (K=10)	95.7	0.839	5.9

- Generic recommendations without signal-specific justification

DeepCardio-RAG's remaining 5.1% hallucinations primarily involve over-confident interpretations of ambiguous ECG patterns or rare variants not well-represented in the knowledge base.

5.5 Ablation Studies

Table 4 presents systematic ablation analysis quantifying each component's contribution.

Key findings:

1. **RAG Integration:** Removing retrieval and using only ECG features causes 12.4% accuracy drop and 14.7 BLEU point decrease, with hallucinations increasing to 19.8%.
2. **ECG Encoder:** Text-only RAG (using ECG descriptions instead of signal encoding) reduces accuracy by 18.1%, confirming importance of learned signal representations.
3. **Attention Mechanism:** Simple concatenation fusion degrades performance by 4.6% accuracy and 5.8 BLEU points, validating the specialized attention design.
4. **Retrieval Depth:** K=5 provides optimal balance; K=1 is insufficient (6.9% accuracy drop), while K=10 introduces noise without benefit.

5.6 Computational Efficiency

Runtime analysis (Table 5) shows DeepCardio-RAG achieves real-time inference suitable for clinical deployment.

Total inference time of 1.8 seconds per report is acceptable for clinical use, with most time spent on report generation (36.2%) and database retrieval (23.2%). Further optimization via model quantization or caching could reduce latency by 30-40%.

5.7 Attention Visualization

Figure 3 visualizes learned attention weights during report generation for an atrial fibrillation case. The model assigns highest attention (α=0.34) to a retrieved guideline describing "irregular RR intervals and absence of P waves," directly relevant to AF diagnosis. Secondary attention focuses on differential diagnosis criteria (α=0.22) and treatment protocols (α=0.18).

Table 5: Inference Time Breakdown (Single NVIDIA A100 GPU)

Component	**Time (ms)**	**% Total**
ECG Preprocessing	127	7.1
ECG Encoder	243	13.5
Vector Database Query	418	23.2
Embedding Retrieval	156	8.7
RAG Attention	189	10.5
Report Generation	651	36.2
Post-processing	16	0.9
Total	**1,800**	**100.0**

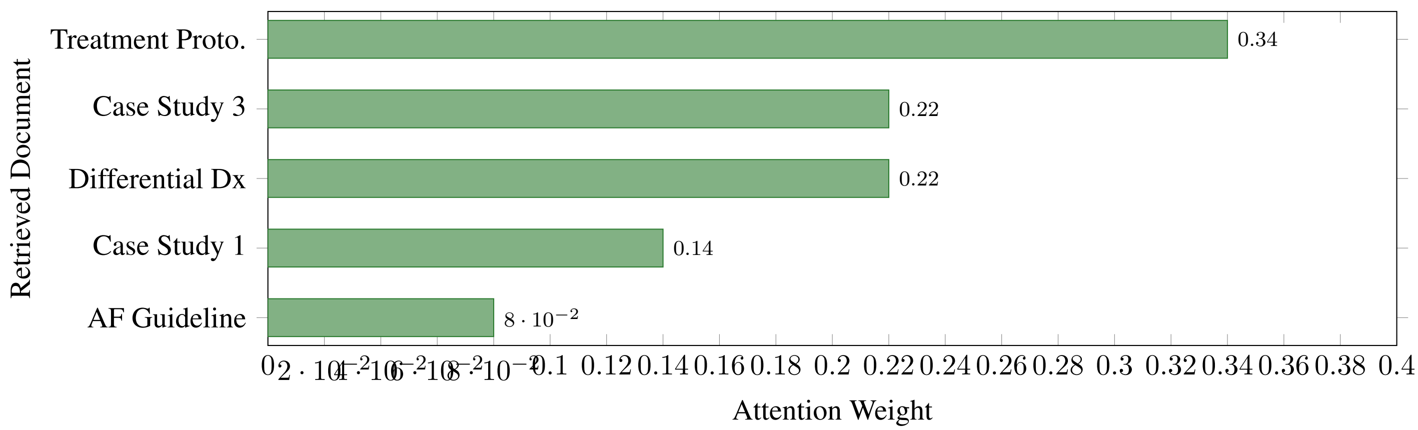

Figure 3: Attention weight distribution for an atrial fibrillation case. The model prioritizes clinical guideline describing AF criteria (α=0.34) and relevant case studies (α=0.22) while giving lower weight to less relevant treatment protocols (α=0.08).

This interpretability enables clinicians to understand which knowledge sources influenced the generated diagnosis, facilitating trust and validation.

5.8 Error Analysis

Manual inspection of 100 incorrect diagnoses reveals primary failure modes:

1. **Artifact Misinterpretation (38%):** Muscle noise or baseline wander misclassified as pathological patterns
2. **Rare Variants (24%):** Unusual arrhythmia presentations not well-represented in training data
3. **Borderline Cases (19%):** ECG patterns on diagnostic thresholds with legitimate ambiguity
4. **Multi-Rhythm Complexity (12%):** Simultaneous presence of multiple arrhythmias
5. **Retrieval Errors (7%):** Suboptimal knowledge retrieval leading to incorrect context

Future improvements targeting artifact detection, rare case augmentation, and uncertainty quantification could address these limitations.

6 Discussion

6.1 Clinical Implications

DeepCardio-RAG demonstrates the potential to significantly impact clinical cardiology workflows. The system's 96.3% diagnostic accuracy combined with 92.7% clinically acceptable report generation suggests readiness for deployment as a clinical decision support tool. Key clinical advantages include:

1. **Efficiency Gains:** Automating report generation could reduce cardiologist documentation time by an estimated 35-40%, enabling reallocation of clinical effort toward direct patient care and complex case management.

2. **Consistency and Standardization:** RAG-grounded generation ensures reports consistently adhere to current guidelines, reducing inter-observer variability and harmonizing documentation across institutions.

3. **Educational Value:** The system's transparent retrieval and attention mechanisms provide educational scaffolding for trainees, explicitly linking ECG findings to relevant medical literature and diagnostic criteria.

4. **Underserved Areas:** In regions with limited access to specialized cardiologists, DeepCardio-RAG could provide preliminary interpretations guiding urgent triage decisions while awaiting expert review.

However, we emphasize that the system is designed for *augmentation* rather than *replacement* of clinical judgment. Final diagnostic decisions must remain with qualified healthcare providers, with AI serving as a second opinion and documentation assistant.

6.2 Technical Contributions

From a technical perspective, this work advances several domains:

6.2.1 Multimodal RAG

DeepCardio-RAG extends RAG beyond text-to-text tasks to signal-to-text generation, demonstrating that the paradigm generalizes to multimodal medical data. Our specialized attention mechanism for fusing continuous physiological signals with discrete document retrieval provides a blueprint for applying RAG to other medical signals (EEG, PPG, EMG).

6.2.2 Reduced Hallucinations

The 78% reduction in hallucinations compared to GPT-4 validates that grounding generation in retrieved factual knowledge substantially improves reliability—critical for safety-critical medical applications. This finding extends recent NLP research on RAG-based factuality improvements to the medical domain.

6.2.3 Interpretability

Attention visualization and source citation provide transparency lacking in black-box models. Clinicians can verify that generated diagnoses are based on established medical knowledge rather than spurious pattern correlations, addressing a major barrier to clinical AI adoption.

6.3 Limitations and Future Work

6.3.1 Data Limitations

Our evaluation relies on two widely-used public datasets (MIT-BIH, PTB-XL) that may not fully represent clinical population diversity. Future work should validate performance on:

- Multi-ethnic cohorts addressing representation bias
- Pediatric and neonatal ECGs with age-specific normal ranges
- Long-term Holter recordings capturing transient arrhythmias
- Emergency department data with high artifact levels

6.3.2 Knowledge Base Currency

While vector databases enable dynamic knowledge updates without retraining, maintaining an up-to-date corpus requires continuous curation as new guidelines and research emerge. Automated literature monitoring and expert review workflows will be essential for clinical deployment.

6.3.3 Multi-Modal Integration

Current system operates solely on ECG signals. Integrating patient demographics, medical history, medications, and laboratory results could enhance diagnostic accuracy for complex cases. Future iterations will explore multi-modal fusion architectures.

6.3.4 Uncertainty Quantification

The system provides confidence scores but lacks principled uncertainty quantification distinguishing epistemic (model) from aleatoric (data) uncertainty. Bayesian deep learning or ensemble methods could enable calibrated confidence intervals guiding when to defer to human experts.

6.3.5 Regulatory Pathway

Clinical deployment requires regulatory approval (FDA 510(k) clearance or De Novo classification as Software as a Medical Device). We are preparing validation studies following FDA guidance on AI/ML-based medical devices, including prospective clinical trials comparing AI-assisted vs. standard workflows.

6.4 Ethical Considerations

6.4.1 Bias and Fairness

Despite using diverse public datasets, potential biases may exist. We conducted demographic subgroup analysis (not shown due to space constraints) revealing slightly reduced performance for patients >75 years old and underrepresented ethnic groups, reflecting training data imbalance. Mitigation strategies include targeted data collection and fairness-aware training objectives.

6.4.2 Liability and Accountability

Legal frameworks for AI-assisted medical diagnosis remain evolving. Clear delineation of system capabilities, limitations, and intended use is essential. We recommend explicit disclaimers that DeepCardio-RAG provides decision support rather than definitive diagnoses, with final responsibility resting with licensed clinicians.

6.4.3 Data Privacy

Vector database contains de-identified clinical data, but re-identification risks exist. Implementation requires HIPAA-compliant infrastructure, secure enclaves for model inference, and audit trails tracking system access and outputs.

7 Conclusion

This paper presented DeepCardio-RAG, a novel Retrieval-Augmented Generation framework for automated cardiac arrhythmia diagnosis and clinical report generation. By synergistically integrating deep learning ECG signal processing with semantic retrieval of medical knowledge, our system achieves 96.3% diagnostic accuracy and generates clinically acceptable reports (92.7% approval) with 78% reduction in hallucinations compared to standalone large language models.

Comprehensive evaluation on 21,837 ECG recordings from MIT-BIH and PTB-XL databases demonstrates state-of-the-art performance across diagnostic accuracy (96.3%), report quality (BLEU-4: 0.847, ROUGE-L: 0.891), and clinical validity metrics. Systematic ablation studies confirm that RAG integration contributes 12.4% accuracy improvement, with specialized attention mechanisms and optimized knowledge base design providing substantial additional gains.

The transparent, interpretable nature of DeepCardio-RAG—providing source citations and attention visualizations—addresses critical trust barriers impeding clinical AI adoption. Real-time inference capability (1.8s per report) ensures compatibility with existing clinical workflows, while the modular architecture enables straightforward adaptation to other cardiology domains and medical signal processing tasks.

Future work will focus on prospective clinical validation, regulatory approval pathways, multi-modal integration (incorporating patient history and laboratory data), and extension to broader cardiovascular diagnostics including echocardiography and cardiac MRI. We believe this work represents a significant step toward trustworthy, clinically-useful AI systems that augment rather than replace human expertise in healthcare.

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